

DELFT UNIVERSITY OF TECHNOLOGY

COMPUTATIONAL TRANSPORT PHENOMENA
CH3421

Assignment 5

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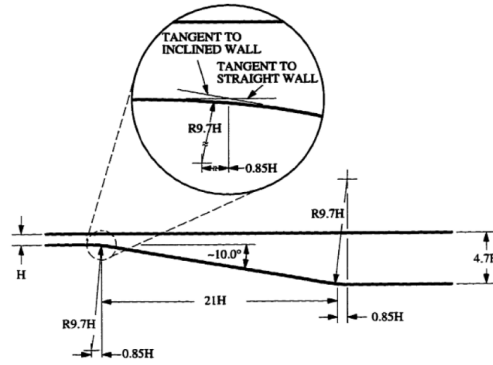


Figure 1: Asymmetric diffuser geometry

Flow separation is often met in many engineering problems including diffusers, turbomachinery and aircrafts. What is interesting, the flow separation and reattachment is still a challenging task for many CFD codes.

In this study we will consider a well-known validation case of asymmetric two-dimensional diffuser, also known as a Buice-Eaton 2D diffuser. The flow through such a diffuser includes three important features:

- Well defined inlet conditions (fully-developed, turbulent inlet flow)
- Smooth-wall separation due to an adverse pressure gradient
- Reattachment of the boundary layer

We will model two-dimensional, incompressible, turbulent, fully-developed flow with Reynolds number of 20,000. The geometry of the asymmetric diffuser is presented in Fig. 1. The x-axis is taken along the main flow direction and y-axis in straight-wall normal direction.

Solve the problem using the Standard $k-\varepsilon$ model and the Standard $k-\omega$ model and compare the results with the experimental solution of Buice and Eaton [1].

Step-by-step guide to the report

- Write a short introduction (2-3 lines) to the problem
- Show the geometry you are considering and the equations you are going to solve
- Explain how you set up the numerical solution of the problem, showing the mesh and the FLUENT settings used (models, boundary conditions, solution controls, initialization)
- Show a snapshot of your residuals and of the wall shear over the iterations
- Show your results in terms of streamlines
- Show your results in terms of velocity profiles and discuss how they compare between each others and with the experimental data

References

- [1] C. U. Buice and J. K. Eaton. “Experimental Investigation of Flow Through an Asymmetric Plane Diffuser”. In: *Journal of Fluids Engineering* 122.June (2000), pp. 433–435.